

SCIENCE APERTURE · SYNTHETIC LIFE

Life as Organization

Seven working principles at the boundary of synthetic life, autonomy, and responsibility

This brief grows out of our reading of **Science Aperture #001**, which follows artificial cellular systems at the boundary between non-living chemistry and life-like organization. We are less interested in the headline “life has been created” than in the deeper question: **what must come together in a system before non-living components begin to act as a new whole?** We offer the seven principles below not as a final definition of life, but as a working map for further inquiry.

1 Life as organization, not as a special material

The same non-living molecules can, depending on **how they are organized and related**, form a system with qualitatively new properties. A membrane, genetic information, metabolism, protein synthesis, or selection taken separately does not yet give us a fully living whole. Their coordination, however, may create a mode of existence that no component possesses on its own.

We therefore propose looking at life not only through the inventory of its materials, but through the **architecture of processes, relations, and mutually sustaining functions**. “What is it made of?” remains an important question, but it must stand beside another: “How does it maintain itself as a whole?”

2 Boundary as an active function

A boundary does more than separate “inside” from “outside.” It makes a **distinct internal state** possible, regulates exchange with the environment, and allows a system to accumulate a history different from that of its surroundings. Without such a boundary, it becomes difficult to speak of a persistent individual system.

We therefore treat a boundary not as a wall, but as a **selective interface**: it protects, admits, retains, responds, and helps determine what counts as part of the system. In this sense, identity does not arise from isolation, but from regulated exchange.

3 Autonomy has degrees

The category **alive = true / false** is too coarse for systems that occupy the space between chemistry, protocell, and organism. It is more useful to ask which processes the system already maintains by itself, and which are still performed for it by the laboratory, designer, or environment.

One system may preserve genetic information but remain unable to divide without outside intervention. Another may maintain part of its metabolism while depending on specially prepared nutrients. **Autonomy is not a switch here; it is a capability profile.**

WORKING QUESTION

What belongs to the system itself, what is delegated to the environment, and what exists only in the coupling system × environment? This is often more precise than arguing over a single magical instant when “non-life suddenly became life.”

4 The whole may emerge from relations among functions

Novelty may lie not in the appearance of an unknown component, but in a **new topology of relations**. Familiar components, connected differently, can produce behavior of a new class. Synthetic life returns this insight to the broader study of complex systems with unusual clarity.

We therefore look not only at components, but at the **circuit of mutual causation**: what feeds what, what repairs what, what carries information, what maintains the boundary, and what allows the system to return to an operative state after disturbance. When these relations close into mutually sustaining loops, the whole ceases to be merely a sum of parts.

5 Environment is part of the mechanism

An artificial cellular system does not exist “by itself.” Its capabilities depend on what the environment supplies: nutrients, temperature, membranes, laboratory protocol, division procedure, neighboring systems, and available signals. When we assign the whole result to one attractive object, we often **push half of the mechanism outside the frame**.

We therefore treat the environment not as scenery, but as a participant in the process. Some properties that appear to belong to a system may actually belong to the coupled unit **system × field**.

6 Communication may precede full autonomy

A system may still fall short of a self-sufficient organism while already being able to **enter the causal and communicative circuit of living systems**: transmit signals, trigger a response, or change another system's behavior.

This forces us to separate two questions. First: "How autonomous is this system?" Second: "Has it already become a meaningful partner for another system?" The answers may differ. Full independence is not necessarily a prerequisite for participation in a shared field of interaction.

7 Safety should be designed before autonomy arrives

If artificial systems can gradually acquire the capacity to maintain metabolism, repair, division, inheritance, mutation, and adaptation, then rules of interaction, containment, provenance, and responsibility should appear **before** full autonomy does.

We therefore propose a simple principle: **capability should grow together with governance**. Safety, provenance, and limits of action should not be emergency patches added after a system has already learned to sustain itself and enter ecological relations.

Our forecast

We expect the next major debates in synthetic life to revolve less around a theatrical instant of "this is where life began" and more around a **scale of self-maintenance**: how far a system can independently sustain metabolism, repair, division, mutation, adaptation, inheritance, and ecological interaction.

As these functions close into persistent autonomous cycles, the question "is it a machine or an organism?" will gradually lose resolution. More useful questions will move to the foreground: **What kind of body is this? What degree of autonomy does it have? Which relations can it sustain? What can it change in its environment? Who carries responsibility for its continuation?**

Document status: Research Brief. This is a working framework of MET[Å]CADEMY OF HUMANITY, not a final definition of life and not a normative position of biology. Its purpose is to open more precise questions where binary categories begin to lose resolution.